

Forms of a complex number

Basic skills

Q1.

a) $z = 6 + 8i$

$$|z| = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$$

$$\text{Arg } z = \tan^{-1}\left(\frac{8}{6}\right) = 0.927$$

b) $z = -5 + 6i$

$$|z| = \sqrt{(-5)^2 + 6^2} = \sqrt{61}$$

$$\text{Arg } z = \tan^{-1}\left(\frac{6}{-5}\right) = -0.876 + \pi$$

AS $x < 0$
 $y > 0$
↙

c) $z = 3 - i$

$$|z| = \sqrt{3^2 + (-1)^2} = \sqrt{10}$$

$$\text{Arg } z = \tan^{-1}\left(-\frac{1}{3}\right) = -0.322$$

Q2. a) $z = 1 + i$

$$|z| = \sqrt{1^2 + 1^2} = \sqrt{2} \quad \text{Arg } z = \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4}$$

b) so mod-arg form $z = |z| (\cos \text{Arg } z + i \sin \text{Arg } z)$

$$z = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

c) $\cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$, $\sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$ so yes

d) exponential form $z = |z| e^{i \text{Arg } z}$

$$z = \sqrt{2} e^{\frac{\pi}{4} i}$$

Competency skills

Q 3. for $z = 5 - i$

$$|z| = \sqrt{5^2 + 1^2} = \sqrt{26}$$

$$\text{Arg } z = \tan^{-1}\left(\frac{-1}{5}\right) = -0.197$$

so $z = \sqrt{26} (\cos(-0.197) + i \sin(-0.197))$

$$z = \sqrt{26} e^{-0.197 i}$$

for $z = 2 \cos \frac{2\pi}{3} + 2i \sin \frac{2\pi}{3}$

As $\cos \frac{2\pi}{3} = -\frac{1}{2}$, $\sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2}$

$$z = 2 \times -\frac{1}{2} + 2i \times \frac{\sqrt{3}}{2}$$
$$= -1 + \sqrt{3}i$$

Also $|z| = 2$, $\text{Arg } z = \frac{2\pi}{3}$

So $z = 2e^{\frac{2\pi}{3}i}$

for $z = \sqrt{3}e^{-\frac{3\pi}{4}i}$

$$|z| = \sqrt{3}, \text{Arg } z = -\frac{3\pi}{4}$$

$$z = \sqrt{3} \left(\cos\left(-\frac{3\pi}{4}\right) + i \sin\left(-\frac{3\pi}{4}\right) \right)$$

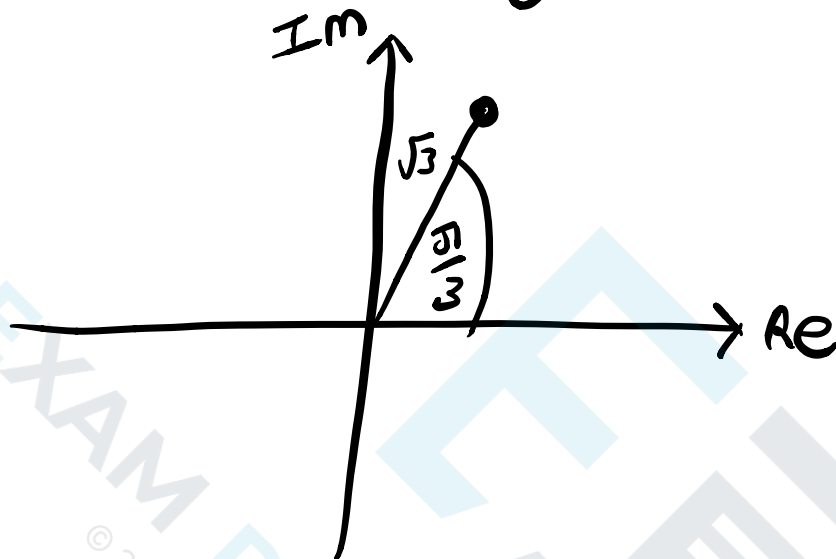
As $\cos\left(-\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2}$, $\sin\left(-\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2}$

$$z = \sqrt{3} \times -\frac{\sqrt{2}}{2} + \sqrt{3} \times -\frac{\sqrt{2}}{2} i$$

$$= -\frac{\sqrt{6}}{2} - \frac{\sqrt{6}}{2} i$$

Q 4. $z = \sqrt{3} \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$

$|z| = \sqrt{3}$ $\text{Arg } z = \frac{\pi}{3}$



Q 5.

$z = 5e^{\frac{3\pi}{7}i}$

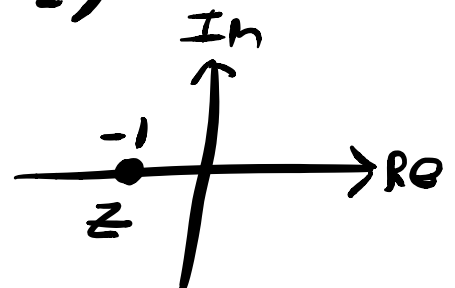
$|z| = 5$, $\text{Arg } z = \frac{3\pi}{7}$



Advanced skills

Q 6. $z = \cos \pi + i \cos \left(-\frac{\pi}{2} \right)$

$= -1 + 0i = -1$



Q 7.

$$z = |z| e^{i \operatorname{Arg} z}$$

$$= |z| (\cos(\operatorname{Arg} z) + i \sin(\operatorname{Arg} z))$$

$$z^* = |z| (\cos(\operatorname{Arg} z) - i \sin(\operatorname{Arg} z))$$

$$= |z| (\cos(-\operatorname{Arg} z) + i \sin(-\operatorname{Arg} z))$$

$$= |z| e^{-i \operatorname{Arg} z}$$

Q 8.

$$z = 2(1-i)$$

$$|z| = 2\sqrt{2}$$

$$\operatorname{Arg} z = \tan^{-1}(-1) = -\frac{\pi}{4}$$

$$z = 2\sqrt{2} e^{-\frac{\pi}{4}i}$$

$$w = 2i$$

$$|w| = 2$$

$$\operatorname{Arg} w = \frac{\pi}{2}$$

$$w = 2 e^{\frac{\pi}{2}i}$$

$$\begin{aligned} \text{a) } z^2 &= (2\sqrt{2} e^{-\frac{\pi}{4}i})^2 = 4 \times 2 \times e^{-\frac{\pi}{2}i} \\ &= 8 e^{-\frac{\pi}{2}i} \end{aligned}$$

$$\begin{aligned} \text{b) } z^2 w^3 &= 8 e^{-\frac{\pi}{2}i} \times (2 e^{\frac{\pi}{2}i})^3 \\ &= 8 e^{-\frac{\pi}{2}i} \times 8 e^{\frac{3\pi}{2}i} = 64 e^{\pi i} \end{aligned}$$

$$c) \quad \frac{z}{2w} = \frac{\cancel{2}\sqrt{2}e^{-\frac{\pi}{4}i}}{\cancel{2} \times 2e^{\frac{\pi}{2}i}} = \frac{\sqrt{2}}{2} e^{-\frac{3\pi}{4}i}$$

$$d) \quad \frac{z}{2e^{0.5}} = \frac{\cancel{2}\sqrt{2}e^{-\frac{\pi}{4}i}}{\cancel{2}e^{0.5}} = \sqrt{2}e^{-\frac{\pi}{4}i - 0.5}$$

$$= \sqrt{2}e^{-0.5}e^{-\frac{\pi}{4}i}$$